DUAL-DAMPING LINEARIZED ADMM FOR NONSMOOTH COMPOSITE OPTIMIZATION

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Abstract. We study a class of structured nonconvex constrained optimization problems with nonsmooth composite terms. We develop a dual-damping linearized augmented Lagrangian method and analyze its approximate KKT residual guarantees.

Key words. augmented Lagrangian method, nonsmooth optimization, ADMM, inearized methods, KKT residual

AMS subject classifications. 65K05, 90C26, 90C30, 90C46

1. Introduction.

In this paper, we consider the following structured optimization problem

$$(\mathbf{P}) \qquad \min_{\boldsymbol{x}\in\mathbb{R}^n} \quad F(\boldsymbol{x}) = f(\boldsymbol{x}) + h(\mathcal{A}(\boldsymbol{x}))$$

where $f\colon \mathbb{R}^n \to \mathbb{R}$ is a smooth (Lipschitz differentiable) function, and h is a proper closed prox-friendly function that is continuous on its closed domain. The mapping $\mathcal{A}\colon \mathbb{R}^n \to \mathbb{R}^m$ is continuously differentiable on $\mathbb{R}^n$ with Lipschitz continuous Jacobian; the complete regularity conditions are deferred to Section 2.1. Problems of the form (**P**) arise naturally in data science, machine learning, and statistics: f serves as a smooth data-fidelity term, while h is a nonsmooth regularizer that promotes desired structural properties in $\mathcal{A}(\boldsymbol{x})$ [6, 13, 14].

For h being the indicator function of a closed convex set, problem (**P**) reduces to a standard nonlinear programming problem with nonlinear constraints. In this case, the augmented Lagrangian method (ALM) is a classical and powerful framework for solving (**P**) and has been extensively studied in the literature; see, e.g., [2, 15] for comprehensive treatments. For the general problem (**P**), the ALM framework can be adapted by introducing an auxiliary variable $\boldsymbol{y} = \mathcal{A}(\boldsymbol{x})$ and reformulating (**P**) as a constrained optimization problem with linear constraints. The resulting ALM-type algorithms typically involve solving a sequence of augmented Lagrangian subproblems, which can be computationally expensive in practice. To mitigate this issue, linearized ALM-type algorithms have been developed, which linearize the smooth part of the augmented Lagrangian and add a proximal term to the primal update; see, e.g., [9, 10, 17] for recent developments in this direction. The best-known complexity guarantees for these algorithms are $O(\epsilon^{-3})$ for finding an ϵ-approximate KKT point of (**P**) under suitable assumptions; see, e.g., [9, 10, 17].

Single-loop algorithms do not require that the subproblems be solved to the required accuracy or that the number of inner iterations be tuned to improve the solution accuracy, and thus are often simpler to implement in practice. The study of single-loop algorithms has received much attention recently, and many algorithms of this kind have been proposed to solve (**P**) under various structural assumptions. A notable common feature is that in most of these studies, h is either Lipschitz or the indicator function of a closed convex set; see [3, 8, 18, 4, 7, 5, 16], which assumed h to be Lipschitz

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continuous, and [1, 7, 16, 11, 12], which studied the case where h is the indicator function of a closed convex set. However, the case where h is a general non-Lipschitz function is not covered by existing single-loop algorithms.

The main contributions are as follows.

- We derive DDLALM from a dual-damping reformulation whose damped multiplier admits a closed-form expression. The resulting algorithm is single-loop and uses an explicit first-order primal update.
- We prove boundedness of the anchor multiplier under a damped anchor update and establish a reduced-potential descent mechanism. The smoothness constants depend on the bounded anchor multiplier $\boldsymbol{\mu}^k$, rather than on an a priori bound for the virtual damped multiplier $\boldsymbol{\lambda}^k$.
- Under compact working-set and uniform LICQ assumptions, we derive a weighted finite-length estimate for the primal steps. This estimate is the key link between the scheduled effective penalty and the KKT residual.
- With $s_k \asymp k^{1/3}$ and $\eta_k \asymp k^{-1/3}$, we show that a selected iterate satisfies

$$\left\|\nabla f(\boldsymbol{x}^\kappa) + J_{\mathcal{A}}(\boldsymbol{x}^\kappa)^\top \boldsymbol{y}^\kappa\right\| = O(K^{-1/3}), \qquad \|\mathcal{A}(\boldsymbol{x}^\kappa)\| = O(K^{-1/3}),$$

  which yields an $O(\epsilon^{-3})$ iteration complexity for finding an approximate KKT point.

2. Preliminaries. Notation. We use $\langle\cdot,\cdot\rangle$ and $\|\cdot\|$ for the Euclidean inner product and norm. For a matrix, $\|\cdot\|$ denotes the spectral norm. For $\mathcal{A} = (A_1, \ldots, A_m)^\top$, let $J_{\mathcal{A}}(\boldsymbol{x}) \in \mathbb{R}^{m\times n}$ denote its Jacobian, whose ith row is $\nabla A_i(\boldsymbol{x})^\top$. We write $\sigma_{\min}(J_{\mathcal{A}}(\boldsymbol{x}))$ for the smallest singular value of $J_{\mathcal{A}}(\boldsymbol{x})$.

For a proper closed convex function h, its convex subdifferential and proximal mapping are, respectively,

$$\partial h(y) := \{v : \ h(z) \geq h(y) + \langle v, z - y\rangle \text{ for all } z\},$$

$$\operatorname{prox}_{\gamma h}(u) := \operatorname*{argmin}_{y\in\mathbb{R}^m} \left\{ h(y) + \frac{1}{2\gamma}\|y - u\|^2 \right\}, \qquad \gamma > 0. \tag{2.1}$$

We define the corresponding KKT system of Problem (**P**) as

$$\begin{aligned} \nabla f(x) + J_{\mathcal{A}}(x)^\top \lambda &= \mathbf{0}, \\ \lambda &\in \partial h(y), \\ r(x, y) &= \mathbf{0}. \end{aligned}$$

Definition 2.1 (ϵ-KKT point). *For $\epsilon > 0$, the triple (x, y, λ) is called an ϵ-KKT point of (3.1) if there exists $\boldsymbol{\lambda} \in \partial h(y)$ such that*

$$\begin{cases} \|\nabla f(x) + J_{\mathcal{A}}(x)^\top \lambda\| \leq \epsilon, \\ \|\mathcal{A}(x) - y\| \leq \epsilon. \end{cases} \tag{2.2}$$

2.1. Assumptions.

Assumption 1 (Level-bounded objective). Assume that F is level-bounded, that is, for every $\alpha \in \mathbb{R}$, the set

$$\mathcal{S}_\alpha := \{\boldsymbol{x} \in \mathbb{R}^n : F(\boldsymbol{x}) \leq \alpha\}$$

is empty or compact.

Assumption 2. There exists a compact convex set $\mathcal{S} \subseteq \mathbb{R}^n$ containing the generated sequence $\{\boldsymbol{x}^k\}_{k\geq 0}$ such that the following conditions hold:

(i) The function $f : \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable, and there exists $L_f \geq 0$ such that

$$\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{x}')\| \leq L_f \|\boldsymbol{x} - \boldsymbol{x}'\|, \qquad \forall \boldsymbol{x}, \boldsymbol{x}' \in \mathcal{S}.$$

(ii) The mapping $\mathcal{A} : \mathbb{R}^n \to \mathbb{R}^m$ is continuously differentiable, and there exist $L_A, G_A \geq 0$ such that

$$\|J_{\mathcal{A}}(\boldsymbol{x}) - J_{\mathcal{A}}(\boldsymbol{x}')\| \leq L_A \|\boldsymbol{x} - \boldsymbol{x}'\|, \qquad \forall \boldsymbol{x}, \boldsymbol{x}' \in \mathcal{S},$$

and

$$\|J_{\mathcal{A}}(\boldsymbol{x})\| \leq G_A, \qquad \forall \boldsymbol{x} \in \mathcal{S}.$$

Since $\mathcal{A}$ is continuous and $\mathcal{S}$ is compact, the constant

$$B_A := \sup_{\boldsymbol{x}\in\mathcal{S}} \|\mathcal{A}(\boldsymbol{x})\| < \infty.$$

Assumption 3. The function $h : \mathbb{R}^m \to \mathbb{R}$ is proper, closed, convex, and globally L_h-Lipschitz. Consequently,

$$\|\boldsymbol{\xi}\| \leq L_h, \qquad \forall \boldsymbol{y} \in \mathbb{R}^m, \quad \forall \boldsymbol{\xi} \in \partial h(\boldsymbol{y}). \tag{2.3}$$

3. Dual-Damping Linearized Augmented Lagrangian Method. This section develops a single-loop dual-damping linearized augmented Lagrangian method for problem (**P**). The method keeps the augmented-Lagrangian penalty fixed, introduces a proximal regularization on the dual variable, and applies a Gauss–Seidel update in the order $\boldsymbol{x}$-step, $\boldsymbol{y}$-step, and multiplier steps.

3.1. Dual-Damping Reformulation. Introduce the auxiliary variable $\boldsymbol{y} = \mathcal{A}(\boldsymbol{x})$. Problem (**P**) is equivalently written as

$$\min_{\boldsymbol{x}\in\mathbb{R}^n,\, \boldsymbol{y}\in\mathbb{R}^m} \; f(\boldsymbol{x}) + h(\boldsymbol{y}) \quad \text{s.t.} \quad \mathcal{A}(\boldsymbol{x}) - \boldsymbol{y} = \boldsymbol{0}. \tag{3.1}$$

For a fixed penalty $\rho > 0$, a dual-regularizer parameter $s > 0$, and an anchor multiplier $\boldsymbol{\mu} \in \mathbb{R}^m$, define

$$\mathcal{P}_s(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\lambda}; \boldsymbol{\mu}) := f(\boldsymbol{x}) + h(\boldsymbol{y}) + \langle \boldsymbol{\lambda}, \mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\rangle + \frac{\rho}{2}\|\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\|^2 - \frac{1}{2s}\|\boldsymbol{\lambda} - \boldsymbol{\mu}\|^2. \tag{3.2}$$

For fixed $(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\mu})$, the map $\boldsymbol{\lambda} \mapsto \mathcal{P}_s(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\lambda}; \boldsymbol{\mu})$ is $1/s$-strongly concave. Its unique maximizer is

$$\boldsymbol{\lambda}_s(\boldsymbol{x}, \boldsymbol{y}; \boldsymbol{\mu}) = \boldsymbol{\mu} + s\big(\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\big). \tag{3.3}$$

Substituting (3.3) into (3.2) gives the reduced dual-damping augmented Lagrangian

$$\Phi_s(\boldsymbol{x}, \boldsymbol{y}; \boldsymbol{\mu}) := f(\boldsymbol{x}) + h(\boldsymbol{y}) + \langle \boldsymbol{\mu}, \mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\rangle + \frac{\bar{\rho}_s}{2}\|\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\|^2, \qquad \bar{\rho}_s := \rho + s. \tag{3.4}$$

For later use, fix $(\boldsymbol{x}, \boldsymbol{\mu})$ and minimize (3.4) with respect to $\boldsymbol{y}$. The resulting proximal point is

$$\boldsymbol{y}_s(\boldsymbol{x}; \boldsymbol{\mu}) = \operatorname{prox}_{h/\bar{\rho}_s}\left(\mathcal{A}(\boldsymbol{x}) + \frac{\boldsymbol{\mu}}{\bar{\rho}_s}\right). \tag{3.5}$$

3.2. Algorithm Design. Let $\{s_k\}_{k\geq 0}$ be a positive dual-damping sequence and define

$$\bar{\rho}_k := \rho + s_k. \tag{3.6}$$

where $\{s_k\}_{k\geq 0}$ is nondecreasing. At iteration k, the method starts from $(\boldsymbol{x}^k, \boldsymbol{y}^k, \boldsymbol{\mu}^k)$. Define the current feasibility residual and the multiplier driving the $\boldsymbol{x}$-step by

$$\boldsymbol{r}^k := \mathcal{A}(\boldsymbol{x}^k) - \boldsymbol{y}^k, \qquad \boldsymbol{w}^k := \boldsymbol{\mu}^k + \bar{\rho}_k \boldsymbol{r}^k. \tag{3.7}$$

Gradient descent $\boldsymbol{x}$-step. The first block is updated by linearizing the smooth part of $\Phi_{s_k}(\cdot, \boldsymbol{y}^k; \boldsymbol{\mu}^k)$ at $\boldsymbol{x}^k$:

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - \eta_k \left[\nabla f(\boldsymbol{x}^k) + J_{\mathcal{A}}(\boldsymbol{x}^k)^\top \boldsymbol{w}^k\right]. \tag{3.8}$$

Exact $\boldsymbol{y}$-step. With $\boldsymbol{x}^{k+1}$ fixed, the second block is updated by exactly minimizing $\Phi_{s_k}(\boldsymbol{x}^{k+1}, \cdot; \boldsymbol{\mu}^k)$:

$$\boldsymbol{y}^{k+1} = \operatorname{prox}_{h/\bar{\rho}_k}\left(\mathcal{A}(\boldsymbol{x}^{k+1}) + \frac{\boldsymbol{\mu}^k}{\bar{\rho}_k}\right). \tag{3.9}$$

Exact $\boldsymbol{\lambda}$-step. We update $\boldsymbol{\lambda}$ by using an exact update.

$$\boldsymbol{\lambda}^{k+1} := \boldsymbol{\mu}^k + s_k(\mathcal{A}(\boldsymbol{x}^{k+1}) - \boldsymbol{y}^{k+1}). \tag{3.10}$$

Gradient ascent $\boldsymbol{\mu}$-step. We perform the update of μ^{k+1} as the gradient ascent:

$$\boldsymbol{\mu}^{k+1} = \left(1 - \frac{\tau_k}{s_k}\right)\boldsymbol{\mu}^k + \frac{\tau_k}{s_k}\boldsymbol{\lambda}^{k+1}, \tag{3.11}$$

where the stepsize of the dual variable μ^k satisfies $0 < \tau_k \leq s_k$. The complete method is summarized in Algorithm 3.1.

Algorithm 3.1 Dual-Damping Linearized Augmented Lagrangian Method (DDLALM)

1: **Input:** $\boldsymbol{x}^0 \in \mathbb{R}^n$, $\boldsymbol{y}^0 \in \operatorname{dom} h$, and $\boldsymbol{\mu}^0 \in \mathbb{R}^m$, fixed base penalty $\rho > 0$, sequences $\{s_k\}$, $\{\eta_k\}$, and $\{\tau_k\}$, and iteration limit K.
2: **for** $k = 0, 1, \ldots, K-1$ **do**
3: Update $\boldsymbol{x}^{k+1}$ by (3.8).
4: Update $\boldsymbol{y}^{k+1}$ by (3.9).
5: Update $\boldsymbol{\lambda}^{k+1}$ by (3.10)
6: Update $\boldsymbol{\mu}^{k+1}$ by (3.11).
7: **end for**

4. Auxiliary Lemmas. This section establishes the basic descent properties of the DDLALM algorithm.

For convenience, we introduce the successive differences

$$\Delta_{\boldsymbol{x}}^{k+1} := \boldsymbol{x}^{k+1} - \boldsymbol{x}^k, \qquad \Delta_{\boldsymbol{y}}^{k+1} := \boldsymbol{y}^{k+1} - \boldsymbol{y}^k, \qquad \Delta_{\boldsymbol{\mu}}^{k+1} := \boldsymbol{\mu}^{k+1} - \boldsymbol{\mu}^k.$$

and set $\bar{\rho}_k := \rho + s_k$.

Define

$$M_\mu := \max\{\|\boldsymbol{\mu}^0\|, L_h\}, \qquad C_r := L_h + M_\mu,$$

and introduce the splitting residual and the KKT-aligned multiplier

$$\boldsymbol{r}^{k+1} := \mathcal{A}(\boldsymbol{x}^{k+1}) - \boldsymbol{y}^{k+1}, \qquad \boldsymbol{z}^{k+1} := \boldsymbol{\mu}^k + \bar{\rho}_k \boldsymbol{r}^{k+1}.$$

Subtracting $\boldsymbol{\mu}^k$ from (3.11) and using (3.10) gives the identity

$$\Delta_{\boldsymbol{\mu}}^{k+1} = \tau_k \boldsymbol{r}^{k+1}, \qquad k \geq 0. \tag{4.1}$$

LEMMA 4.1 (Multiplier and splitting-residual bounds). *Suppose that Assumption 3 holds. Let $\{(\boldsymbol{x}^k, \boldsymbol{y}^k, \boldsymbol{\mu}^k)\}_{k\geq 0}$ be generated by Algorithm 3.1. Then, for every $k \geq 0$,*

$$\|\boldsymbol{\mu}^k\| \leq M_\mu, \qquad \|\boldsymbol{r}^{k+1}\| \leq \frac{C_r}{\bar{\rho}_k}, \qquad \|\Delta_{\boldsymbol{\mu}}^{k+1}\| \leq \frac{\tau_k C_r}{\bar{\rho}_k}. \tag{4.2}$$

Proof. The first-order optimality condition of the exact $\boldsymbol{y}$-update is

$$\mathbf{0} \in \partial h(\boldsymbol{y}^{k+1}) - \boldsymbol{\mu}^k - \bar{\rho}_k \boldsymbol{r}^{k+1},$$

which proves $\boldsymbol{z}^{k+1} \in \partial h(\boldsymbol{y}^{k+1})$. By (2.3), $\|\boldsymbol{z}^{k+1}\| \leq L_h$.

The identity $\boldsymbol{r}^{k+1} = (\boldsymbol{z}^{k+1} - \boldsymbol{\mu}^k)/\bar{\rho}_k$ and (4.1) yield

$$\boldsymbol{\mu}^{k+1} = (1 - q_k)\boldsymbol{\mu}^k + q_k \boldsymbol{z}^{k+1}, \qquad q_k := \frac{\tau_k}{\bar{\rho}_k} \in (0, 1), \tag{4.3}$$

which is a convex combination. Induction gives $\|\boldsymbol{\mu}^k\| \leq M_\mu$ for every $k \geq 0$.

The definition of $\boldsymbol{r}^{k+1}$ now gives

$$\|\boldsymbol{r}^{k+1}\| \leq \frac{\|\boldsymbol{z}^{k+1}\| + \|\boldsymbol{\mu}^k\|}{\bar{\rho}_k} \leq \frac{C_r}{\bar{\rho}_k},$$

which proves the residual estimate in (4.2). Combining this estimate with (4.1) gives the increment estimate, completing the proof. ∎

The lemma below characterizes the smoothness of $\Phi_s(\boldsymbol{x}, \boldsymbol{y}; \boldsymbol{\mu})$ with respect to $\boldsymbol{x}$ for fixed $\boldsymbol{\mu}$ and $\boldsymbol{y}$.

LEMMA 4.2. *Suppose that Assumption 2 and the conditions of Lemma 4.1 hold. Define*

$$B_y := \max\left\{\|\boldsymbol{y}^0\|, B_A + \frac{M_\mu + L_h}{\bar{\rho}_0}\right\}.$$

Then $\|\boldsymbol{y}^k\| \leq B_y$ for every $k \geq 0$. Moreover, for every $k \geq 0$ and every $\boldsymbol{x}, \boldsymbol{x}' \in \mathcal{S}$,

$$\begin{gathered}\left\|\nabla_{\boldsymbol{x}} \Phi_{s_k}(\boldsymbol{x}, \boldsymbol{y}^k; \boldsymbol{\mu}^k) - \nabla_{\boldsymbol{x}} \Phi_{s_k}(\boldsymbol{x}', \boldsymbol{y}^k; \boldsymbol{\mu}^k)\right\| \leq L_k \|\boldsymbol{x} - \boldsymbol{x}'\|, \\ L_k := L_f + L_A M_\mu + \bar{\rho}_k \left[L_A(B_A + B_y) + G_A^2\right].\end{gathered} \tag{4.4}$$

In particular,

$$L_k \leq \widehat{L}\bar{\rho}_k, \qquad \widehat{L} := \frac{L_f + L_A M_\mu}{\bar{\rho}_0} + L_A(B_A + B_y) + G_A^2. \tag{4.5}$$

Proof. Set

$$\boldsymbol{u}^{k+1} := \mathcal{A}(\boldsymbol{x}^{k+1}) + \frac{\boldsymbol{\mu}^k}{\bar{\rho}_k}.$$

The optimality condition of the proximal update (3.9) gives

$$\bar{\rho}_k(\boldsymbol{u}^{k+1} - \boldsymbol{y}^{k+1}) \in \partial h(\boldsymbol{y}^{k+1}).$$

Hence (2.3), the multiplier bound, and $\boldsymbol{x}^{k+1} \in \mathcal{S}$ imply

$$\begin{aligned}\|\boldsymbol{y}^{k+1}\| &\le \|\mathcal{A}(\boldsymbol{x}^{k+1})\| + \frac{\|\boldsymbol{\mu}^k\| + L_h}{\bar{\rho}_k} \\ &\le B_A + \frac{M_\mu + L_h}{\bar{\rho}_0} \le B_y,\end{aligned}$$

where the second inequality uses the monotonicity of $\{\bar{\rho}_k\}$. The definition of B_y also covers $\boldsymbol{y}^0$.

Fix $k \ge 0$ and $\boldsymbol{x}, \boldsymbol{x}' \in \mathcal{S}$. From (3.4),

$$\nabla_{\boldsymbol{x}} \Phi_{s_k}(\boldsymbol{x}, \boldsymbol{y}^k; \boldsymbol{\mu}^k) = \nabla f(\boldsymbol{x}) + J_{\mathcal{A}}(\boldsymbol{x})^\top(\boldsymbol{\mu}^k - \bar{\rho}_k \boldsymbol{y}^k) + \bar{\rho}_k J_{\mathcal{A}}(\boldsymbol{x})^\top \mathcal{A}(\boldsymbol{x}).$$

Because $\mathcal{S}$ is convex, the Jacobian bound integrated along the segment joining $\boldsymbol{x}'$ and $\boldsymbol{x}$ gives

$$\|\mathcal{A}(\boldsymbol{x}) - \mathcal{A}(\boldsymbol{x}')\| \le G_A \|\boldsymbol{x} - \boldsymbol{x}'\|.$$

Moreover,

$$\begin{aligned}&J_{\mathcal{A}}(\boldsymbol{x})^\top \mathcal{A}(\boldsymbol{x}) - J_{\mathcal{A}}(\boldsymbol{x}')^\top \mathcal{A}(\boldsymbol{x}') \\ &= \big(J_{\mathcal{A}}(\boldsymbol{x}) - J_{\mathcal{A}}(\boldsymbol{x}')\big)^\top \mathcal{A}(\boldsymbol{x}) + J_{\mathcal{A}}(\boldsymbol{x}')^\top \big(\mathcal{A}(\boldsymbol{x}) - \mathcal{A}(\boldsymbol{x}')\big),\end{aligned}$$

and therefore

$$\big\|J_{\mathcal{A}}(\boldsymbol{x})^\top \mathcal{A}(\boldsymbol{x}) - J_{\mathcal{A}}(\boldsymbol{x}')^\top \mathcal{A}(\boldsymbol{x}')\big\| \le (L_A B_A + G_A^2)\|\boldsymbol{x} - \boldsymbol{x}'\|.$$

Combining these estimates with $\|\boldsymbol{\mu}^k - \bar{\rho}_k \boldsymbol{y}^k\| \le M_\mu + \bar{\rho}_k B_y$ proves (4.4). Finally, $\bar{\rho}_k \ge \bar{\rho}_0$ gives (4.5). □

The next lemma gives a quasi-descent estimate for the potential function under the stepsize condition.

LEMMA 4.3. *Suppose that Assumptions 2 and 3 hold and that the conditions of Lemma 4.2 are satisfied. Let $\theta_x > 0$ satisfy $\theta_x \widehat{L} \le 1/2$, and choose*

$$\eta_k = \frac{\theta_x}{\bar{\rho}_k}, \qquad k \ge 0. \tag{4.6}$$

Then, for every $k \ge 0$, it holds that

$$\begin{aligned}&\Phi_{s_{k+1}}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}; \boldsymbol{\mu}^{k+1}) - \Phi_{s_k}(\boldsymbol{x}^k, \boldsymbol{y}^k; \boldsymbol{\mu}^k) \\ &\quad \le -\frac{3\bar{\rho}_k}{4\theta_x}\|\Delta_{\boldsymbol{x}}^{k+1}\|^2 - \frac{\bar{\rho}_k}{2}\|\Delta_{\boldsymbol{y}}^{k+1}\|^2 + \left(\frac{1}{\tau_k} + \frac{\bar{\rho}_{k+1} - \bar{\rho}_k}{2\tau_k^2}\right)\|\Delta_{\boldsymbol{\mu}}^{k+1}\|^2.\end{aligned} \tag{4.7}$$

Proof. We decompose the one-step change into two parts: the primal descent produced by the $(\boldsymbol{x}, \boldsymbol{y})$-updates, and the dual ascent produced by the $\boldsymbol{\mu}$-update together with the penalty-parameter drift.

For the primal descent. Since $\mathcal{S}$ is convex and contains $\boldsymbol{x}^k$ and $\boldsymbol{x}^{k+1}$, the estimate in Lemma 4.2 applies along the segment joining these two points. Moreover,

$$\nabla_{\boldsymbol{x}} \Phi_{s_k}(\boldsymbol{x}^k, \boldsymbol{y}^k; \boldsymbol{\mu}^k) = \nabla f(\boldsymbol{x}^k) + J_{\mathcal{A}}(\boldsymbol{x}^k)^\top \boldsymbol{w}^k,$$

where $\boldsymbol{w}^k := \boldsymbol{\mu}^k + \bar{\rho}_k \boldsymbol{r}^k$. The $\boldsymbol{x}$-update (3.8) gives the identity

$$\left\langle \nabla_{\boldsymbol{x}} \Phi_{s_k}(\boldsymbol{x}^k, \boldsymbol{y}^k; \boldsymbol{\mu}^k), \Delta_{\boldsymbol{x}}^{k+1} \right\rangle = -\frac{1}{\eta_k} \|\Delta_{\boldsymbol{x}}^{k+1}\|^2,$$

so the integral form of the descent lemma gives

$$\Phi_{s_k}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^k; \boldsymbol{\mu}^k) - \Phi_{s_k}(\boldsymbol{x}^k, \boldsymbol{y}^k; \boldsymbol{\mu}^k) \le -\left(\frac{1}{\eta_k} - \frac{L_k}{2}\right) \|\Delta_{\boldsymbol{x}}^{k+1}\|^2.$$

Since h is convex, the exact $\boldsymbol{y}$-minimization (3.9) gives

$$\Phi_{s_k}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}; \boldsymbol{\mu}^k) - \Phi_{s_k}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^k; \boldsymbol{\mu}^k) \le -\frac{\bar{\rho}_k}{2} \|\Delta_{\boldsymbol{y}}^{k+1}\|^2.$$

Both steps hold $(\boldsymbol{\mu}^k, \bar{\rho}_k)$ fixed, so adding them yields the primal bound

$$\Phi_{s_k}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}; \boldsymbol{\mu}^k) - \Phi_{s_k}(\boldsymbol{x}^k, \boldsymbol{y}^k; \boldsymbol{\mu}^k) \le -\left(\frac{1}{\eta_k} - \frac{L_k}{2}\right) \|\Delta_{\boldsymbol{x}}^{k+1}\|^2 - \frac{\bar{\rho}_k}{2} \|\Delta_{\boldsymbol{y}}^{k+1}\|^2.$$

For the dual ascent. Holding $(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1})$ fixed, the $\boldsymbol{\mu}$-update and the change of the penalty coefficient from $\bar{\rho}_k$ to $\bar{\rho}_{k+1}$ give

$$\begin{aligned}
&\Phi_{s_{k+1}}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}; \boldsymbol{\mu}^{k+1}) - \Phi_{s_k}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}; \boldsymbol{\mu}^k) \\
&\quad = \langle \Delta_{\boldsymbol{\mu}}^{k+1}, \boldsymbol{r}^{k+1} \rangle + \frac{\bar{\rho}_{k+1} - \bar{\rho}_k}{2} \|\boldsymbol{r}^{k+1}\|^2 \\
&\quad = \tau_k \|\boldsymbol{r}^{k+1}\|^2 + \frac{\bar{\rho}_{k+1} - \bar{\rho}_k}{2} \|\boldsymbol{r}^{k+1}\|^2 \\
&\quad = \left(\frac{1}{\tau_k} + \frac{\bar{\rho}_{k+1} - \bar{\rho}_k}{2\tau_k^2}\right) \|\Delta_{\boldsymbol{\mu}}^{k+1}\|^2,
\end{aligned}$$

where the second equality uses (4.1), i.e. $\Delta_{\boldsymbol{\mu}}^{k+1} = \tau_k \boldsymbol{r}^{k+1}$, and the last equality substitutes $\boldsymbol{r}^{k+1} = \Delta_{\boldsymbol{\mu}}^{k+1}/\tau_k$.

Combining the two parts yields

$$\begin{aligned}
&\Phi_{s_{k+1}}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}; \boldsymbol{\mu}^{k+1}) - \Phi_{s_k}(\boldsymbol{x}^k, \boldsymbol{y}^k; \boldsymbol{\mu}^k) \\
&\qquad \le -\left(\frac{1}{\eta_k} - \frac{L_k}{2}\right) \|\Delta_{\boldsymbol{x}}^{k+1}\|^2 - \frac{\bar{\rho}_k}{2} \|\Delta_{\boldsymbol{y}}^{k+1}\|^2 \\
&\qquad\quad + \left(\frac{1}{\tau_k} + \frac{\bar{\rho}_{k+1} - \bar{\rho}_k}{2\tau_k^2}\right) \|\Delta_{\boldsymbol{\mu}}^{k+1}\|^2.
\end{aligned} \tag{4.8}$$

Moreover,

$$\frac{1}{\eta_k} - \frac{L_k}{2} \geq \frac{3}{4\eta_k} = \frac{3\bar{\rho}_k}{4\theta_x}. \tag{4.9}$$

The inequality follows from (4.5), (4.6), and $\theta_x \widehat{L} \leq 1/2$. Substituting it into (4.8) proves (4.7). ∎

PROPOSITION 4.4 (Potential upper bound and boundedness consequences). *Suppose that Assumption 1 and the conditions of Lemma 4.3 hold. Assume that*

$$C_e := C_r^2 \sum_{k=0}^{\infty} \left(\frac{\tau_k}{\bar{\rho}_k^2} + \frac{\bar{\rho}_{k+1} - \bar{\rho}_k}{2\bar{\rho}_k^2} \right) < \infty. \tag{4.10}$$

Define

$$\overline{\Phi} := \Phi_{s_0}(\boldsymbol{x}^0, \boldsymbol{y}^0; \boldsymbol{\mu}^0) + C_e, \qquad \alpha := \overline{\Phi} + \frac{C_r^2}{2\bar{\rho}_0}. \tag{4.11}$$

Then, for every $k \geq 0$,

$$\Phi_{s_k}(\boldsymbol{x}^k, \boldsymbol{y}^k; \boldsymbol{\mu}^k) \leq \overline{\Phi}, \qquad F(\boldsymbol{x}^k) \leq \alpha. \tag{4.12}$$

Consequently, define

$$S := \mathcal{S}_\alpha := \{\boldsymbol{x} \in \mathbb{R}^n : F(\boldsymbol{x}) \leq \alpha\}. \tag{4.13}$$

Then S is compact and

$$\boldsymbol{x}^k \in S, \qquad k \geq 0, \tag{4.14}$$

Moreover, Lemma 4.2 gives $\|\boldsymbol{y}^k\| \leq B_y$ for every $k \geq 0$. Thus $\{\boldsymbol{x}^k\}$, $\{\boldsymbol{y}^k\}$, and $\{\boldsymbol{\mu}^k\}$ are bounded. For later use, the constants

$$G_f := \sup_{\boldsymbol{x} \in S} \|\nabla f(\boldsymbol{x})\|, \qquad C_A := \sup_{\boldsymbol{x} \in S} \|\mathcal{A}(\boldsymbol{x})\| \tag{4.15}$$

are finite.

Proof. Applying (4.2) to the last term in (4.7), dropping the nonpositive primal terms, and summing from $j = 0$ to $k - 1$ give

$$\Phi_{s_k}(\boldsymbol{x}^k, \boldsymbol{y}^k; \boldsymbol{\mu}^k) \leq \Phi_{s_0}(\boldsymbol{x}^0, \boldsymbol{y}^0; \boldsymbol{\mu}^0) + C_e = \overline{\Phi}.$$

Let $\boldsymbol{r}^k := \mathcal{A}(\boldsymbol{x}^k) - \boldsymbol{y}^k$. The global Lipschitz continuity of h and the multiplier bound $\|\boldsymbol{\mu}^k\| \leq M_\mu$ yield

$$\begin{aligned} F(\boldsymbol{x}^k) &\leq f(\boldsymbol{x}^k) + h(\boldsymbol{y}^k) + L_h \|\boldsymbol{r}^k\| \\ &= \Phi_{s_k}(\boldsymbol{x}^k, \boldsymbol{y}^k; \boldsymbol{\mu}^k) - \langle \boldsymbol{\mu}^k, \boldsymbol{r}^k \rangle - \frac{\bar{\rho}_k}{2} \|\boldsymbol{r}^k\|^2 + L_h \|\boldsymbol{r}^k\| \\ &\leq \Phi_{s_k}(\boldsymbol{x}^k, \boldsymbol{y}^k; \boldsymbol{\mu}^k) + C_r \|\boldsymbol{r}^k\| - \frac{\bar{\rho}_k}{2} \|\boldsymbol{r}^k\|^2 \\ &\leq \overline{\Phi} + \frac{C_r^2}{2\bar{\rho}_k} \leq \alpha. \end{aligned}$$

Assumption 1 proves that $\mathcal{S}_\alpha$ is compact, and (4.12) gives $\boldsymbol{x}^k \in \mathcal{S}_\alpha = S$, which proves (4.14). The bound on $\{\boldsymbol{y}^k\}$ follows from Lemma 4.2, and boundedness of $\{\boldsymbol{\mu}^k\}$ follows from Lemma 4.1. Finally, continuity of ∇f and $\mathcal{A}$ on the compact set S proves (4.15). ∎

PROPOSITION 4.5 (Weighted finite length). *Suppose that Assumption 1 and the conditions of Lemma 4.3 hold. Assume in addition that*

$$\bar{\rho}_k = \rho + c_s(k+1)^a, \qquad c_s > 0, \quad a \in (0,1).$$

Assume also that the dual stepsizes satisfy

$$\sum_{k=0}^{\infty} \frac{\tau_k}{\bar{\rho}_k^2} < \infty. \tag{4.16}$$

Then there exists a constant $C_{\mathrm{desc}} > 0$, independent of K, such that, for every $K \geq 1$,

$$\sum_{k=0}^{K-1} \bar{\rho}_k \left(\|\Delta_{\boldsymbol{x}}^{k+1}\|^2 + \|\Delta_{\boldsymbol{y}}^{k+1}\|^2 \right) \leq C_{\mathrm{desc}}. \tag{4.17}$$

Proof. Throughout the proof, $C > 0$ denotes a generic constant independent of k and K. The value of C may change from line to line and depends only on the problem data and fixed algorithmic parameters.

Step 1: Summability of the controlled ascent error. Define

$$e_k := C_r^2 \left(\frac{\tau_k}{\bar{\rho}_k^2} + \frac{\bar{\rho}_{k+1} - \bar{\rho}_k}{2\bar{\rho}_k^2} \right).$$

The first series in e_k is finite by (4.16). For the penalty-drift series, the concavity of $t \mapsto t^a$ and $\bar{\rho}_k = \rho + c_s(k+1)^a$ give

$$\bar{\rho}_{k+1} - \bar{\rho}_k = c_s\big((k+2)^a - (k+1)^a\big) \leq c_s a (k+1)^{a-1}.$$

Moreover, $\bar{\rho}_k \geq c_s(k+1)^a$, so

$$\frac{\bar{\rho}_{k+1} - \bar{\rho}_k}{\bar{\rho}_k^2} \leq \frac{c_s a (k+1)^{a-1}}{c_s^2 (k+1)^{2a}} = O(k^{-(1+a)}).$$

Consequently, $\sum_{k=0}^{\infty} e_k < \infty$. Proposition 4.4 now applies. Hence the compact set $S = \mathcal{S}_\alpha$ defined in (4.13) contains every $\boldsymbol{x}^k$.

Step 2: Telescoping descent. In (4.7), the ascent term $\left(\frac{1}{\tau_k} + \frac{\bar{\rho}_{k+1}-\bar{\rho}_k}{2\tau_k^2}\right)\|\Delta_{\boldsymbol{\mu}}^{k+1}\|^2$ rewrites as $\tau_k\|\boldsymbol{r}^{k+1}\|^2 + \frac{\bar{\rho}_{k+1}-\bar{\rho}_k}{2}\|\boldsymbol{r}^{k+1}\|^2$ via $\Delta_{\boldsymbol{\mu}}^{k+1} = \tau_k \boldsymbol{r}^{k+1}$, and is bounded by e_k using (4.2). By (4.7), set $c_{\mathrm{desc}} := \min\{3/(4\theta_x), 1/2\} > 0$. Then

$$\Phi_{s_{k+1}}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}; \boldsymbol{\mu}^{k+1}) - \Phi_{s_k}(\boldsymbol{x}^k, \boldsymbol{y}^k; \boldsymbol{\mu}^k) \leq -c_{\mathrm{desc}} \bar{\rho}_k \left(\|\Delta_{\boldsymbol{x}}^{k+1}\|^2 + \|\Delta_{\boldsymbol{y}}^{k+1}\|^2 \right) + e_k, \tag{4.18}$$

Summing (4.18) from $k = 0$ to $K-1$ yields

$$c_{\mathrm{desc}} \sum_{k=0}^{K-1} \bar{\rho}_k \left(\|\Delta_{\boldsymbol{x}}^{k+1}\|^2 + \|\Delta_{\boldsymbol{y}}^{k+1}\|^2 \right) \leq \Phi_{s_0}(\boldsymbol{x}^0, \boldsymbol{y}^0; \boldsymbol{\mu}^0) - \Phi_{s_K}(\boldsymbol{x}^K, \boldsymbol{y}^K; \boldsymbol{\mu}^K) + \sum_{k=0}^{K-1} e_k.$$

Step 3: Lower bound for the potential. Since h is globally L_h-Lipschitz and $\boldsymbol{r}^K = \mathcal{A}(\boldsymbol{x}^K) - \boldsymbol{y}^K$,

$$h(\boldsymbol{y}^K) \geq h\big(\mathcal{A}(\boldsymbol{x}^K)\big) - L_h \|\boldsymbol{r}^K\|.$$

Using this inequality, $\|\boldsymbol{\mu}^K\| \le M_\mu$, and completing the square give

$$\begin{aligned}\Phi_{s_K}(\boldsymbol{x}^K, \boldsymbol{y}^K; \boldsymbol{\mu}^K) &= f(\boldsymbol{x}^K) + h(\boldsymbol{y}^K) + \langle \boldsymbol{\mu}^K, \boldsymbol{r}^K \rangle + \frac{\bar{\rho}_K}{2}\|\boldsymbol{r}^K\|^2 \\ &\ge F(\boldsymbol{x}^K) - C_r\|\boldsymbol{r}^K\| + \frac{\bar{\rho}_K}{2}\|\boldsymbol{r}^K\|^2 \\ &\ge \inf_{\boldsymbol{x}\in S} F(\boldsymbol{x}) - \frac{C_r^2}{2\bar{\rho}_K} \\ &\ge \inf_{\boldsymbol{x}\in S} F(\boldsymbol{x}) - \frac{C_r^2}{2\bar{\rho}_0}.\end{aligned}$$

The infimum is finite because F is continuous on the compact set S. Therefore,

$$\sum_{k=0}^{K-1} \bar{\rho}_k \left(\|\Delta_{\boldsymbol{x}}^{k+1}\|^2 + \|\Delta_{\boldsymbol{y}}^{k+1}\|^2\right) \le \frac{1}{c_{\mathrm{desc}}}\left[\Phi_{s_0}(\boldsymbol{x}^0, \boldsymbol{y}^0; \boldsymbol{\mu}^0) - \inf_{\boldsymbol{x}\in S} F(\boldsymbol{x}) + \frac{C_r^2}{2\bar{\rho}_0} + \sum_{k=0}^{\infty} e_k\right] =: C_{\mathrm{desc}},$$

where C_{desc} is finite and independent of K. ∎

Remark 4.6 (Dual stepsize design). Lemma 4.1 shows that multiplier boundedness only requires $0 < \tau_k \le s_k$. The additional condition needed for the descent argument is

$$\sum_{k=0}^{\infty} \frac{\tau_k}{\bar{\rho}_k^2} < \infty.$$

For $s_k = c_s(k+1)^a$, a family satisfying $\tau_k = O\big(s_k/(k+1)^q\big)$ meets this requirement whenever $q > 1-a$. If a downstream result additionally requires finite total variation of the anchor multiplier, Lemma 4.1 shows that the stronger condition $\sum_k \tau_k/\bar{\rho}_k < \infty$ is sufficient. The global complexity result below uses $q = 1$, namely $\tau_k = \theta_\mu s_k/(k+2)$. This choice makes the potential-ascent error summable and gives $\mathrm{dist}(\boldsymbol{w}^k, \partial h(\boldsymbol{y}^k)) = O(k^{-1})$; finite total variation of $\{\boldsymbol{\mu}^k\}$ is not needed for that result.

THEOREM 4.7 (Global KKT residual complexity). *Suppose that Assumptions 1, 2, and 3 hold. Let $c_s > 0$, $a \in (0,1)$, $\theta_x > 0$ with $\theta_x \widehat{L} \le 1/2$, and $\theta_\mu \in (0,1]$, and choose*

$$s_k = c_s(k+1)^a, \qquad \bar{\rho}_k = \rho + s_k, \qquad \eta_k = \frac{\theta_x}{\bar{\rho}_k}, \qquad \tau_k = \frac{\theta_\mu s_k}{k+2}, \tag{4.19}$$

where $\widehat{L}$ is the constant in (4.5).

For any $K \ge 2$, choose

$$\kappa \in \operatorname*{argmin}_{k\in\{\lceil K/2\rceil,\dots,K-1\}} \bar{\rho}_k \left(\|\Delta_{\boldsymbol{x}}^{k+1}\|^2 + \|\Delta_{\boldsymbol{y}}^{k+1}\|^2\right).$$

Define the effective KKT multiplier

$$\boldsymbol{z}^\kappa := \boldsymbol{\mu}^{\kappa-1} + \bar{\rho}_{\kappa-1}\boldsymbol{r}^\kappa,$$

where $\boldsymbol{r}^\kappa := \mathcal{A}(\boldsymbol{x}^\kappa) - \boldsymbol{y}^\kappa$. Then

$$\begin{aligned}\left\|\nabla f(\boldsymbol{x}^\kappa) + J_{\mathcal{A}}(\boldsymbol{x}^\kappa)^\top \boldsymbol{z}^\kappa\right\| &= O\Big(K^{-(1-a)/2}\Big), \\ \mathrm{dist}\big(\boldsymbol{z}^\kappa, \partial h(\boldsymbol{y}^\kappa)\big) &= 0, \\ \|\boldsymbol{r}^\kappa\| &= O(K^{-a}).\end{aligned} \tag{4.20}$$

Moreover, $\|\boldsymbol{z}^\kappa\| \le L_h$. *In particular, for the balanced choice* $a = 1/3$,

$$\max\left\{\left\|\nabla f(\boldsymbol{x}^\kappa) + J_{\mathcal{A}}(\boldsymbol{x}^\kappa)^\top \boldsymbol{z}^\kappa\right\|, \operatorname{dist}\big(\boldsymbol{z}^\kappa, \partial h(\boldsymbol{y}^\kappa)\big), \|\boldsymbol{r}^\kappa\|\right\} = O(K^{-1/3}).$$

Consequently, to obtain an ϵ*-approximate KKT point in the sense that*

$$\left\|\nabla f(\boldsymbol{x}^\kappa) + J_{\mathcal{A}}(\boldsymbol{x}^\kappa)^\top \boldsymbol{z}^\kappa\right\| \le \epsilon, \qquad \operatorname{dist}\big(\boldsymbol{z}^\kappa, \partial h(\boldsymbol{y}^\kappa)\big) \le \epsilon, \qquad \|\boldsymbol{r}^\kappa\| \le \epsilon,$$

under the assumptions above, it suffices to take

$$K = O(\epsilon^{-3}).$$

Moreover, the virtual multiplier $\boldsymbol{\lambda}^\kappa = \boldsymbol{\mu}^{\kappa-1} + s_{\kappa-1}\boldsymbol{r}^\kappa$ *and the KKT-aligned multiplier* $\boldsymbol{z}^\kappa = \boldsymbol{\mu}^{\kappa-1} + \bar\rho_{\kappa-1}\boldsymbol{r}^\kappa$ *satisfy*

$$\boldsymbol{z}^\kappa - \boldsymbol{\lambda}^\kappa = \rho \boldsymbol{r}^\kappa,$$

and hence

$$\operatorname{dist}\big(\boldsymbol{\lambda}^\kappa, \partial h(\boldsymbol{y}^\kappa)\big) \le \rho\|\boldsymbol{r}^\kappa\| = O(K^{-a}).$$

The multiplier driving the $\boldsymbol{x}$*-step also satisfies*

$$\operatorname{dist}\big(\boldsymbol{w}^\kappa, \partial h(\boldsymbol{y}^\kappa)\big) = O(K^{-1}).$$

The KKT residual is stated with $\boldsymbol{z}^\kappa$ *because* $\boldsymbol{z}^\kappa \in \partial h(\boldsymbol{y}^\kappa)$ *by the preceding* $\boldsymbol{y}$*-step optimality condition.*

Proof. The primal stepsize in (4.19) satisfies (4.6) and the conditions of Lemma 4.3. Moreover, $0 < \tau_k \le s_k$ and

$$\begin{aligned}\sum_{k=0}^{\infty} \frac{\tau_k}{\bar\rho_k^2} &\le \theta_\mu \sum_{k=0}^{\infty} \frac{1}{(k+2)\bar\rho_k} \\ &\le \frac{\theta_\mu}{c_s} \sum_{k=0}^{\infty} \frac{1}{(k+2)(k+1)^a} < \infty.\end{aligned}$$

Thus all conditions of Proposition 4.5 hold, and that proposition gives

$$\sum_{k=0}^{K-1} \bar\rho_k \left(\|\Delta_{\boldsymbol{x}}^{k+1}\|^2 + \|\Delta_{\boldsymbol{y}}^{k+1}\|^2\right) \le C_{\mathrm{desc}}, \qquad \forall K \ge 1.$$

The index set $\{\lceil K/2\rceil, \dots, K-1\}$ has $\lfloor K/2\rfloor$ elements, and $\lfloor K/2\rfloor \ge K/3$ for every $K \ge 2$. Hence, by the definition of κ,

$$\bar\rho_\kappa \left(\|\Delta_{\boldsymbol{x}}^{\kappa+1}\|^2 + \|\Delta_{\boldsymbol{y}}^{\kappa+1}\|^2\right) \le \frac{3C_{\mathrm{desc}}}{K}.$$

Since $\kappa \in \{\lceil K/2\rceil, \dots, K-1\}$ and $\bar\rho_k = \rho + c_s(k+1)^a$, we have $\bar\rho_\kappa \asymp K^a$. Thus

$$\|\Delta_{\boldsymbol{x}}^{\kappa+1}\| + \|\Delta_{\boldsymbol{y}}^{\kappa+1}\| = O\Big(K^{-(1+a)/2}\Big).$$

Step 1: Feasibility and multiplier residuals. By Lemma 4.1,

$$\|\boldsymbol{r}^\kappa\| \le \frac{C_r}{\bar\rho_{\kappa-1}} = O(K^{-a}).$$

The exact $\boldsymbol{y}$-update optimality condition and (2.3) give

$$\boldsymbol{z}^{\kappa} \in \partial h(\boldsymbol{y}^{\kappa}), \qquad \|\boldsymbol{z}^{\kappa}\| \le L_h,$$

which proves the second residual in (4.20). Moreover, $\boldsymbol{z}^{\kappa} - \boldsymbol{\lambda}^{\kappa} = \rho \boldsymbol{r}^{\kappa}$ gives the stated virtual-multiplier residual.

For every $k \ge 1$, the definitions of $\boldsymbol{w}^k$ and $\boldsymbol{z}^k$, together with (4.1), give

$$\begin{aligned} \boldsymbol{w}^k - \boldsymbol{z}^k &= \boldsymbol{\mu}^k - \boldsymbol{\mu}^{k-1} + \big(\bar{\rho}_k - \bar{\rho}_{k-1}\big)\boldsymbol{r}^k \\ &= \big(\tau_{k-1} + \bar{\rho}_k - \bar{\rho}_{k-1}\big)\boldsymbol{r}^k. \end{aligned}$$

Using (4.2) with index $k-1$ and the schedule (4.19), we obtain

$$\begin{aligned} \|\boldsymbol{w}^k - \boldsymbol{z}^k\| &\le C_r \left(\frac{\tau_{k-1}}{\bar{\rho}_{k-1}} + \frac{\bar{\rho}_k - \bar{\rho}_{k-1}}{\bar{\rho}_{k-1}} \right) \\ &\le C_r \left(\frac{\theta_\mu}{k+1} + \frac{a}{k} \right) = O(k^{-1}), \end{aligned}$$

where the second inequality uses $(k+1)^a - k^a \le a k^{a-1}$. Since the exact $\boldsymbol{y}$-update optimality condition gives $\boldsymbol{z}^k \in \partial h(\boldsymbol{y}^k)$,

$$\operatorname{dist}\big(\boldsymbol{w}^k, \partial h(\boldsymbol{y}^k)\big) \le \|\boldsymbol{w}^k - \boldsymbol{z}^k\| = O(k^{-1}).$$

In particular, $\kappa \ge K/2$ gives $\|\boldsymbol{w}^{\kappa} - \boldsymbol{z}^{\kappa}\| = O(K^{-1})$.

Step 2: Stationarity residual. The $\boldsymbol{x}$-update optimality condition gives

$$\nabla f(\boldsymbol{x}^{\kappa}) + J_{\mathcal{A}}(\boldsymbol{x}^{\kappa})^{\top} \boldsymbol{w}^{\kappa} = -\frac{1}{\eta_{\kappa}} \Delta_{\boldsymbol{x}}^{\kappa+1},$$

where $\boldsymbol{w}^{\kappa} := \boldsymbol{\mu}^{\kappa} + \bar{\rho}_{\kappa} \boldsymbol{r}^{\kappa}$. Consequently,

$$\begin{aligned} \big\|\nabla f(\boldsymbol{x}^{\kappa}) + J_{\mathcal{A}}(\boldsymbol{x}^{\kappa})^{\top} \boldsymbol{z}^{\kappa}\big\| &\le \frac{1}{\eta_{\kappa}} \|\Delta_{\boldsymbol{x}}^{\kappa+1}\| + G_A \|\boldsymbol{w}^{\kappa} - \boldsymbol{z}^{\kappa}\| \\ &= O\Big(K^{-(1-a)/2}\Big) + O(K^{-1}) \\ &= O\Big(K^{-(1-a)/2}\Big). \end{aligned}$$

The two residual rates are balanced by solving $a = (1-a)/2$, which gives $a = 1/3$. For this choice, both stationarity and splitting residuals are $O(K^{-1/3})$, and the iteration complexity claim follows by setting $K = O(\epsilon^{-3})$. ∎

Remark 4.8. The schedule (4.19) couples the dual regularization, primal step, and dual step as

$$\frac{1}{s_k} = \Theta(k^{-a}), \qquad \eta_k = \Theta(k^{-a}), \qquad \tau_k = \Theta(k^{a-1}).$$

The dual-ascent and penalty-drift errors then obey

$$\frac{\tau_k}{\bar{\rho}_k^2} = O(k^{-(1+a)}), \qquad \frac{\bar{\rho}_{k+1} - \bar{\rho}_k}{\bar{\rho}_k^2} = O(k^{-(1+a)}),$$

so both error sequences are summable. In addition,

$$\|\Delta_{\boldsymbol{\mu}}^{k+1}\| = O(k^{-1}), \qquad \operatorname{dist}\big(\boldsymbol{w}^k, \partial h(\boldsymbol{y}^k)\big) = O(k^{-1}).$$

Thus the splitting residual is retained as the feasibility component of the KKT residual, while its contribution to the potential ascent is converted to the summable multiplier-increment term

$$\tau_k \|\boldsymbol{r}^{k+1}\|^2 = \frac{1}{\tau_k}\|\Delta_{\boldsymbol{\mu}}^{k+1}\|^2.$$

Finally, weighted finite length gives

$$r_{\mathrm{stat}} = O(K^{-(1-a)/2}), \qquad r_{\mathrm{feas}} = O(K^{-a}).$$

Balancing these rates yields $a = 1/3$. Hence

$$\frac{1}{s_k} = \Theta(k^{-1/3}), \qquad \eta_k = \Theta(k^{-1/3}), \qquad \tau_k = \Theta(k^{-2/3}),$$

and all three KKT residuals are $O(K^{-1/3})$.

4.1. A conditional local fast rate. The preceding theorem is deliberately stated without local or global error bound assumptions. If one is willing to impose a stronger local condition that rules out nearby infeasible stationary points of the reduced augmented Lagrangian, the same algorithmic template admits a faster local rate with a constant effective penalty. This statement is conditional and should be read as a local refinement rather than as a replacement for Theorem 4.7.

Assumption 4 (Local reduced error bound). There exist an index $k_0 \geq 0$, a constant $\kappa_{\mathrm{eb}} > 0$, and a neighborhood $\mathcal{U}$ of the KKT solution set such that $(\boldsymbol{x}^k, \boldsymbol{\mu}^k) \in \mathcal{U}$ for all $k \geq k_0$, and

$$\|\mathcal{A}(\boldsymbol{x}^k)\| \leq \kappa_{\mathrm{eb}} \left\|\nabla f(\boldsymbol{x}^k) + J_{\mathcal{A}}(\boldsymbol{x}^k)^\top \big(\boldsymbol{\mu}^k + \bar{\rho}\,\mathcal{A}(\boldsymbol{x}^k)\big)\right\|, \qquad \forall k \geq k_0. \tag{4.21}$$

Remark 4.9. Assumption 4 is stronger than uniform LICQ. Uniform LICQ controls multiplier size from a stationarity residual, whereas (4.21) also requires the feasibility violation to be controlled by the stationarity residual of the fixed-penalty reduced model. Equivalently, it excludes local infeasible stationary points of $\Phi(\cdot; \boldsymbol{\mu}^k)$ along the generated anchor sequence.

Corollary 4.10 (Local fast rate under a reduced error bound). *Suppose that the conditions of Proposition 4.4, Assumption* **??**, *and Assumption 4 hold. Let the effective penalty be constant, namely* $\bar{\rho}_k \equiv \bar{\rho} > 0$, *and choose a constant primal stepsize* $\eta > 0$ *such that*

$$\frac{1}{2\eta} - \frac{1}{2}\big(L_f + L_A M_\mu + \bar{\rho}(L_A C_A + G_A^2)\big) \geq \alpha$$

for some $\alpha > 0$. *Assume also that*

$$\sum_{k=0}^{\infty} \frac{\tau_k}{\bar{\rho}} < \infty. \tag{4.22}$$

For any $K > k_0$, *choose*

$$\kappa \in \operatorname*{argmin}_{k \in \{k_0, \ldots, K-1\}} \|\Delta_{\boldsymbol{x}}^{k+1}\|^2,$$

and define

$$\boldsymbol{y}^{\kappa} := \boldsymbol{\mu}^{\kappa} + \bar{\rho}\,\mathcal{A}(\boldsymbol{x}^{\kappa}).$$

Then there exists a constant $C_{\text{fast}} > 0$*, independent of* K*, such that*

$$\left\|\nabla f(\boldsymbol{x}^{\kappa}) + J_{\mathcal{A}}(\boldsymbol{x}^{\kappa})^{\top}\boldsymbol{y}^{\kappa}\right\|^2 + \|\mathcal{A}(\boldsymbol{x}^{\kappa})\|^2 \le \frac{C_{\text{fast}}}{K - k_0}.$$

Consequently, a point satisfying both

$$\left\|\nabla f(\boldsymbol{x}^{\kappa}) + J_{\mathcal{A}}(\boldsymbol{x}^{\kappa})^{\top}\boldsymbol{y}^{\kappa}\right\| \le \epsilon, \qquad \|\mathcal{A}(\boldsymbol{x}^{\kappa})\| \le \epsilon$$

is obtained after $K - k_0 = O(\epsilon^{-2})$ *iterations.*

Proof. With $\bar{\rho}_k \equiv \bar{\rho}$, the penalty-drift term in (4.8) vanishes. The stated stepsize condition gives

$$\Phi(\boldsymbol{x}^{k+1}; \boldsymbol{\mu}^{k+1}) - \Phi(\boldsymbol{x}^{k}; \boldsymbol{\mu}^{k}) \le -\alpha\|\Delta_{\boldsymbol{x}}^{k+1}\|^2 + C_A\|\Delta_{\boldsymbol{\mu}}^{k+1}\|, \qquad \forall k \ge 0,$$

where

$$\Phi(\boldsymbol{x}; \boldsymbol{\mu}) := f(\boldsymbol{x}) + \langle \boldsymbol{\mu}, \mathcal{A}(\boldsymbol{x})\rangle + \frac{\bar{\rho}}{2}\|\mathcal{A}(\boldsymbol{x})\|^2.$$

Summing from $k = k_0$ to $K - 1$, using the lower boundedness of Φ on the compact working set and the summability of $\{\|\Delta_{\boldsymbol{\mu}}^{k+1}\|\}$ implied by (4.2) and (4.22), yields

$$\sum_{k=k_0}^{K-1} \|\Delta_{\boldsymbol{x}}^{k+1}\|^2 \le C$$

for a constant C independent of K. Hence the definition of κ implies

$$\|\Delta_{\boldsymbol{x}}^{\kappa+1}\|^2 \le \frac{C}{K - k_0}.$$

The explicit primal step gives

$$\nabla f(\boldsymbol{x}^{\kappa}) + J_{\mathcal{A}}(\boldsymbol{x}^{\kappa})^{\top}\boldsymbol{y}^{\kappa} = \frac{1}{\eta}(\boldsymbol{x}^{\kappa} - \boldsymbol{x}^{\kappa+1}),$$

so the squared stationarity residual is bounded by $C/(\eta^2(K - k_0))$. Assumption 4 then gives

$$\|\mathcal{A}(\boldsymbol{x}^{\kappa})\|^2 \le \kappa_{\text{eb}}^2 \left\|\nabla f(\boldsymbol{x}^{\kappa}) + J_{\mathcal{A}}(\boldsymbol{x}^{\kappa})^{\top}\boldsymbol{y}^{\kappa}\right\|^2 \le \frac{C}{K - k_0}.$$

Combining the two estimates proves the claim. ∎

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